(Software for)
Modeling and Solving Multiobjective Optimization Problems

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Organization

Part 1 (9:00-10:30):

Fundamentals
Building a complete workflow: from modeling to solving and analyzing

Part 2 (11:00-12:30):

Case Studies
Applying the workflow to increasingly challenging case studies

Material

Resources online:

`https://github.com/xgandibleux/EURO_MCDM2026/`

- Warm-up
- Three case studies

Beforehand, a little warm-up

If not yet done, install *Julia* on your computer

↳ `https://julialang.org/`

1. Discovering the four modes of the REPL

   If not yet done, install the package `IJulia`

2. Editing and running a *Julia* code into *Jupyter* notebook

   If not yet done, install *VScode* and its Julia extension

3. Editing and running a *Julia* code into *VScode*
   Running *Julia* code into the REPL with `include("myProg.jl")`

Case 1

1. If not yet done, install the packages JuMP and HiGHS
2. Coding, solving, and analyzing an IP problem
   ↳ `https://github.com/xgandibleux/EURO_MCDM2026/tree/main/4.case_studies/Case1`

Case 2

1. If not yet done, install the packages MultiObjectiveAlgorithms, Polyhedra, Plots, and PlotlyJS
2. Coding, solving, and analyzing a MOIP problem
   ↳ `https://github.com/xgandibleux/EURO_MCDM2026/tree/main/4.case_studies/Case2`

Case 3

1. If not yet done, install the packages and Pluto
2. Coding, solving, and analyzing an interactive method
   ↳ `https://github.com/xgandibleux/EURO_MCDM2026/blob/main/4.case_studies/Case3/`